Comparative Analysis of Probabilistic and Non-Probabilistic Machine Learning Classifiers for Predicting Stroke Risk Using the Health Stroke Dataset

A Case Study Using Logistic Regression, Naïve Bayes, Support Vector Machine, and K-Nearest Neighbours

1. Introduction

Stroke is one of the leading causes of long-term disability and death worldwide, occurring when blood supply to part of the brain is interrupted or reduced, depriving brain tissue of oxygen and nutrients. According to WHO, millions of people suffer from stroke each year with many cases resulting in permanent physical, cognitive and emotional impairments. Early identification of individuals at high risk of stroke enables timely clinical intervention, lifestyle modification, and preventive care_hence saving lives and reducing the long-term burden on health care systems.

With the growing availability of patient health records, machine learning have improved healthcare by enabling the use of computers to identify patterns and learn relationships between patients characteristics and disease outcomes in clinical data, this can help predict disease outcome and help practitioners make better medical decisions for patients.

This study applies supervised machine learning techniques to the Health Stroke Dataset to predict whether a patient is likely to have a stroke using the two types of supervised classification algorithms: probabilistic classifiers (Logistic Regression and Naïve Bayes), which model class probabilities, and non-probabilistic classifiers (Support Vector Machine and K-Nearest Neighbours), which learn decision boundaries.

The project follows a complete machine learning workflow that includes data preprocessing, model development, hyperparameter tuning, model evaluation, and comparative performance analysis. The findings provide insight into the strengths and limitations of each algorithm and identify the most appropriate model for stroke prediction based on the evaluation metrics.

1.1 Problem Statement

Stroke remains a major global public health challenge because many cases occur unexpectedly despite the presence of identifiable risk factors such as hypertension, age, heart disease, smoking status, body mass index (BMI), and glucose level. Traditional clinical assessment methods may not always capture complex interactions among these variables.

The challenge is therefore to develop an accurate machine learning classification model capable of predicting stroke occurrence from patient health information. Such predictive models can assist healthcare professionals in identifying high-risk individuals earlier, enabling preventive interventions that may reduce the incidence and severity of stroke.

1.2. Aim of the Study

The aim of this study is to compare the predictive performance of probabilistic and non-probabilistic machine learning algorithms for stroke prediction using the Health Stroke Dataset.

1.3 Objectives of the Study

- To preprocess the Health Stroke Dataset for machine learning modelling, including handling missing values, encoding categorical features, and scaling numerical features.
- To develop and tune four classification models: Logistic Regression, Naïve Bayes, Support Vector Machine (SVM), and K-Nearest Neighbours (KNN).
- To evaluate and compare the models using Accuracy, Precision, Recall, and F1-score.
- To plot and interpret ROC curves and compute ROC-AUC for the probabilistic classifiers (Logistic Regression and Naïve Bayes).
- To identify the best-performing model and provide recommendations for stroke-risk prediction.

2. Dataset Description

The Health Stroke Dataset contains 5,110 patient records and 12 colums: a unique patient identifier(ID),demographic attribute(gender, age, ever_married, work_type, Residence_type) and medical information(hypertension, heart_disease, avg_glucose_level, bmi) and lifestyle(smoking_status) collected from patients together with their stroke status. The dataset is intended for binary classification, where the target variable indicates whether a patient has experienced a stroke.

The independent variables describe patient characteristics that have been associated with stroke risk in clinical literature. These include age, gender, hypertension, heart disease, marital status, occupation type, residence type, average glucose level, body mass index (BMI), and smoking status.

The dependent variable is: **Stroke**

- 0 = No Stroke
- 1 = Stroke

Because stroke prediction is a binary classification problem, supervised learning algorithms are appropriate for modelling the relationship between patient characteristics and stroke occurrence.

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3. Data Preprocessing

Data preprocessing is one of the most important stages of any machine learning project because the quality of the input data directly influences model performance. Several preprocessing techniques were applied before model training.

3.1 Importing Libraries and Loading the Dataset

Python libraries including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn were imported to support data manipulation, visualization, preprocessing, model development, and evaluation.

The dataset was loaded into a pandas DataFrame using pandas.read_csv().After loading the dataset, exploratory analysis was performed to examine:

- Number of observations
- Number of features
- Variable names
- Data types
- Statistical summary
- Missing values

This initial inspection provided an understanding of the dataset before preprocessing commenced.The non-informative id column was dropped, and a single record with gender = 'Other' (a rare category with only 1 observation) was removed to avoid unreliable one-hot/label encoding for that category.

3.2 Handling Missing Values

Missing values were identified during exploratory data analysis.Incomplete records can negatively affect machine learning performance by introducing bias and reducing prediction accuracy.The bmi column was found to contain 201 missing values (approximately 3.9% of records), while all other columns were complete. Because BMI is a continuous, right-skewed clinical variable, missing values were imputed using the median BMI computed from the training data's distribution, which is more robust to outliers than the mean.

3.3 Encoding Categorical Variables

Five categorical columns — gender, ever_married, work_type, Residence_type, and smoking_status — were converted into numeric form using Label Encoding, since all downstream models (Logistic Regression, Naïve Bayes, SVM, KNN) require purely numeric input. The resulting encodings were:

- gender: Female = 0, Male = 1
- ever_married: No = 0, Yes = 1
- work_type: Govt_job = 0, Never_worked = 1, Private = 2, Self-employed = 3, children = 4
- Residence_type: Rural = 0, Urban = 1
- smoking_status: Unknown = 0, formerly smoked = 1, never smoked = 2, smokes = 3

3.4 Feature Scaling

The three continuous numerical features — age, avg_glucose_level, and bmi — were standardized using StandardScaler (zero mean, unit variance). This scaling step is essential for SVM, KNN, and Logistic Regression, which are all sensitive to the scale of input features, and does not adversely affect Naïve Bayes. The scaler was fit exclusively on the training set and then applied to the test set to prevent data leakage.

3.5 Train-Test Split

The cleaned and encoded dataset (5,109 records after removing the 'Other' gender row) was split into training (80%) and testing (20%) This produced 4,087 training records and 1,022 test records.Separating the data in this manner reduces the likelihood of overfitting and provides a more reliable estimate of model generalisation on unseen data.

4.0 Model Development

Four classification models were developed: two probabilistic models (Logistic Regression, Naïve Bayes) and two non-probabilistic models (SVM, KNN). For each model, hyperparameters were tuned using 5-fold (3-fold for SVM, for computational efficiency) GridSearchCV, optimizing for F1-score — a metric that balances precision and recall and is more informative than accuracy on this imbalanced dataset.

3.1 Probabilistic Classification Models

Logistic Regression: The Logistic Regression classifier was implemented using Scikit-learn. Hyperparameter tuning was performed to determine the optimal parameter for this dataset. The tuned model obtained from GridSearchCV was used for prediction and evaluation. A grid search was performed over the regularization strength C ∈ {0.01, 0.1, 1, 10, 100} and class_weight ∈ { 'balanced'}, using the L2 penalty and the lbfgs solver.

Naïve Bayes: The Gaussian Naïve Bayes classifier was trained using the preprocessed training dataset. Hyperparameter optimisation was carried out where appropriate to improve predictive performance.Gaussian Naïve Bayes (GaussianNB) was selected because the predictor set consists predominantly of continuous numeric features (age, avg_glucose_level, bmi) alongside binary/encoded categorical features, and GaussianNB assumes each feature is normally distributed within each class. The var_smoothing hyperparameter, which adds a small value to feature variances for numerical stability, was tuned over a logarithmic grid from 1 to 1e-9.

3.2 Non-Probabilistic Classification Models

Support Vector Machine (SVM): finds the hyperplane that maximizes the margin between the two classes. A grid search was performed over C ∈ {0.1, 1, 10}, kernel ∈ {'rbf', 'linear'}, and class_weight ∈ {'balanced'}, with probability=True enabled solely so that ROC-style diagnostics could be inspected (SVM itself is treated here as a non-probabilistic, margin-based classifier).Hyperparameter tuning was performed to determine the most appropriate values for parameters such as:

- C (regularisation parameter)
- Kernel type
- Gamma

The optimal parameter combination identified during GridSearchCV was selected as the final SVM model.

K-**Nearest Neighbours (KNN)**: classifies an observation based on the majority class among its k nearest neighbours in feature space. A grid search was performed over n_neighbors ∈ {3, 5, 7, 9, 11}, weights ∈ {'uniform', 'distance'}, (Manhattan vs. Euclidean distance).The KNN classifier was optimised using GridSearchCV.The tuning process investigated multiple values of:

- Number of neighbours (K)
- Distance metric
- Weighting scheme

The combination that produced the highest validation performance was selected as the final KNN model.

3.3 Selected Hyperparameters

Table 1 summarizes the best hyperparameter combination found for each model.

Model	Selected Hyperparameters (via 5-fold GridSearchCV, scoring = F1)
Logistic Regression	C = 0.1, penalty = 'l2', solver = 'lbfgs', class_weight = 'balanced', max_iter = 1000
Naïve Bayes (Gaussian)	var_smoothing ≈ 1.44 × 10^{-3}
SVM	C = 10, kernel = 'linear', class_weight = 'balanced', probability = True
KNN	n_neighbors = 3, weights = 'distance', p = 1 (Manhattan distance)

Table 1: Best hyperparameters selected via GridSearchCV (scoring = F1-score).

Both Logistic Regression and SVM selected class_weight = 'balanced' as their best configuration, reflecting the fact that the search explicitly rewarded configurations that better handled the 95:5 class imbalance rather than defaulting to the majority class. Naïve Bayes and KNN, which do not expose a class-weighting hyperparameter in the same way, instead relied on their native probability/distance-based mechanisms.

4. Model Evaluation

4.1 Performance Metrics

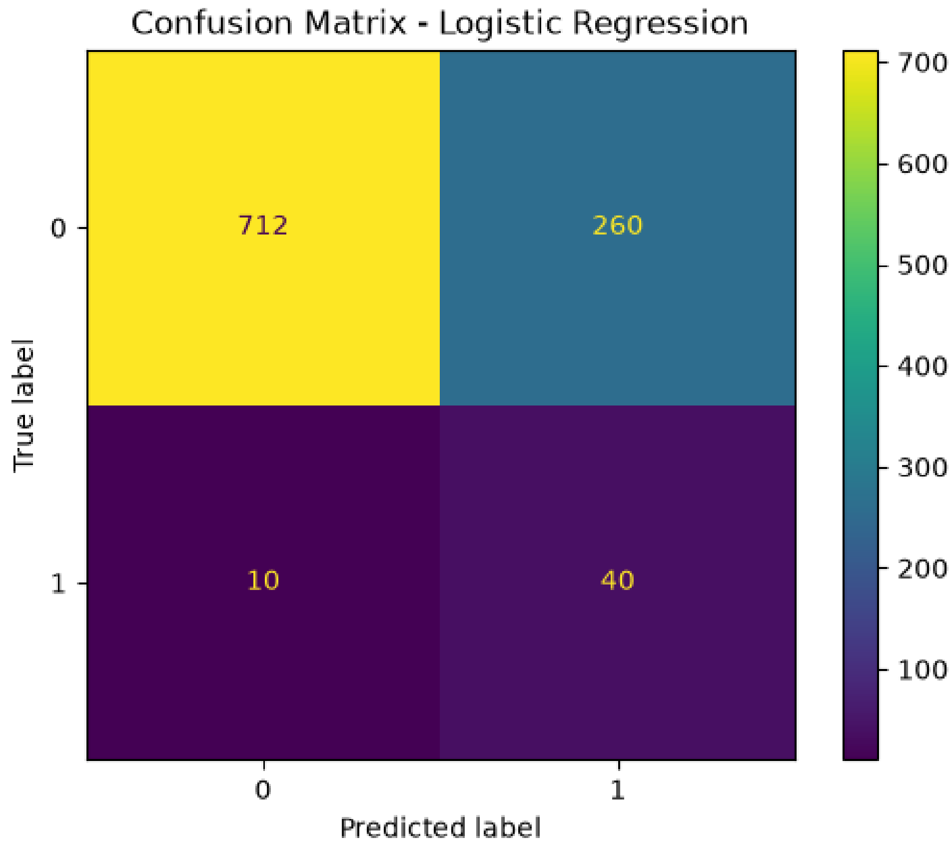
Table 2 reports Accuracy, Precision, Recall, and F1-score (with respect to the positive/stroke class) for all four tuned models on the held-out 20% test set.

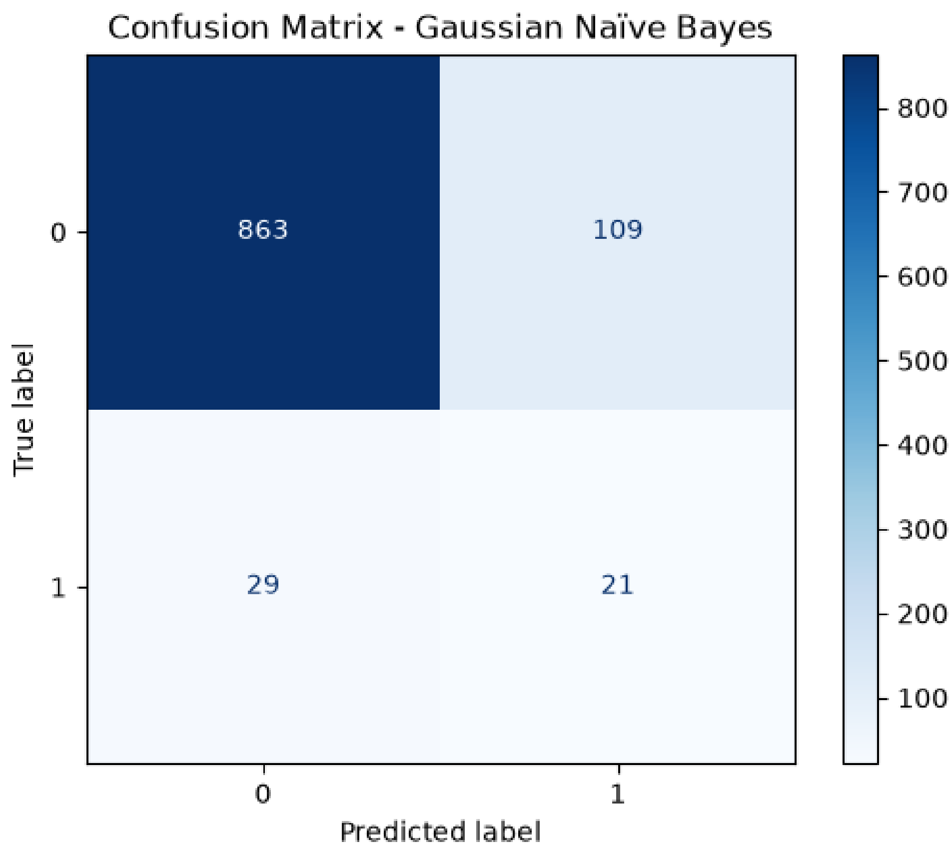
Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.9520	1.000	0.0200	0.0392	0.8426
Naïve Bayes	0.8933	0.1895	0.3600	0.2483	0.8120
SVM	0.7319	0.1603	0.8065	0.2674	
KNN	0.9403	0.6667	0.0323	0.0515	

Table 2: Test-set performance of all four classification models.

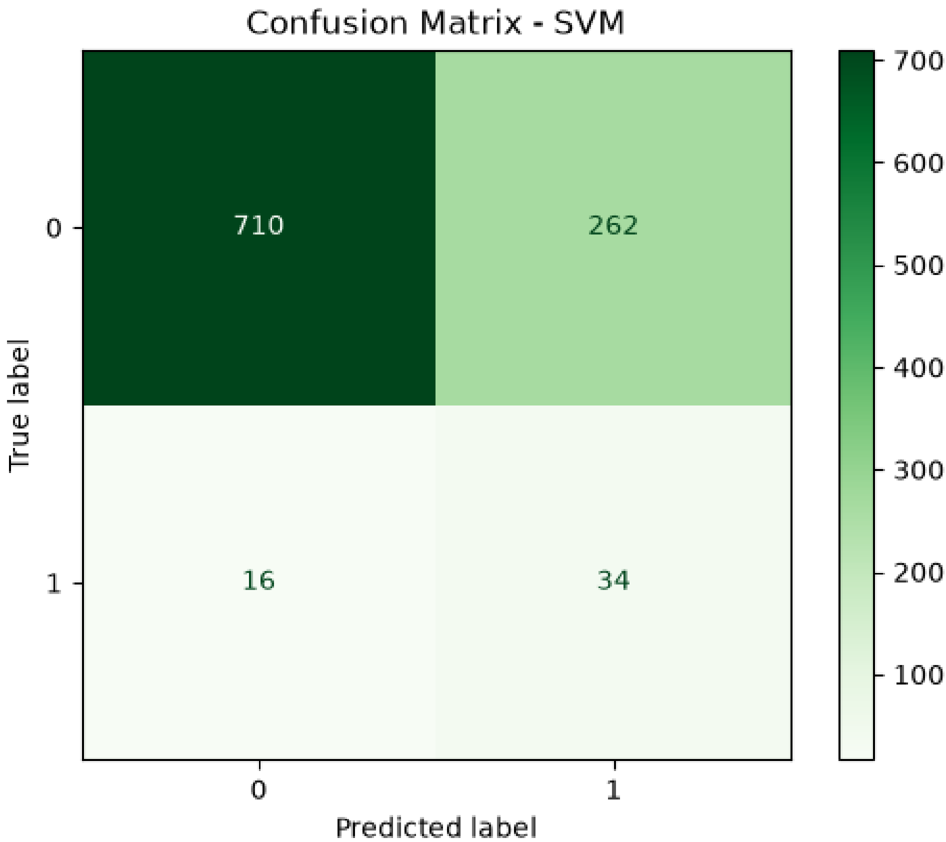
Figure 1: Visual comparison of Accuracy, Precision, Recall, and F1-score across models.

Figure 2: Confusion matrices for all four models on the test set.

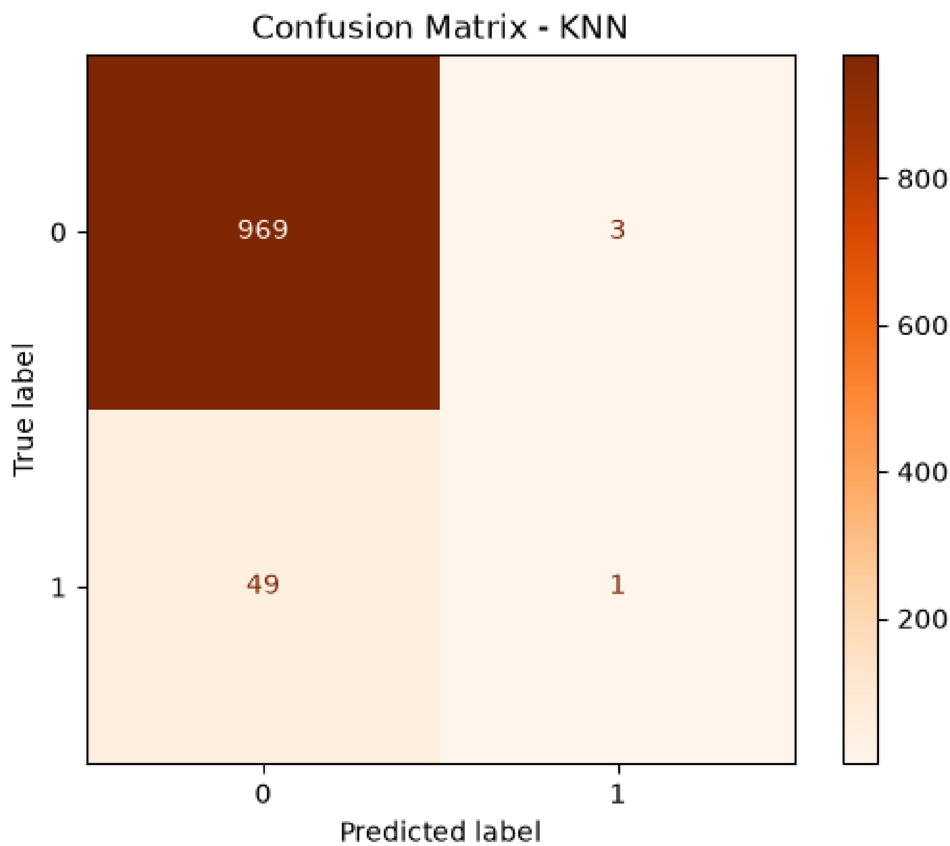

Confusion Matrix - Logistic Regression
True label
Predicted label
0
1
0
1
712
260
10
40
700
600
500
400
300
200
100



Confusion Matrix - Gaussian Naïve Bayes
863
109
29
21
0
1
True label
Predicted label
800
700
600
500
400
300
200
100



Confusion Matrix - SVM
710
262
16
34
True label
Predicted label
0
1
0
1
700
600
500
400
300
200
100



4.2 ROC Curves and AUC (Probabilistic Models)

For the two probabilistic classifiers, Logistic Regression and Naïve Bayes, ROC curves were plotted by varying the classification threshold and computing the True Positive Rate (TPR) and False Positive Rate (FPR) at each threshold. The Area Under the Curve (AUC) summarizes each model's overall ability to rank a randomly chosen stroke patient above a randomly chosen non-stroke patient.

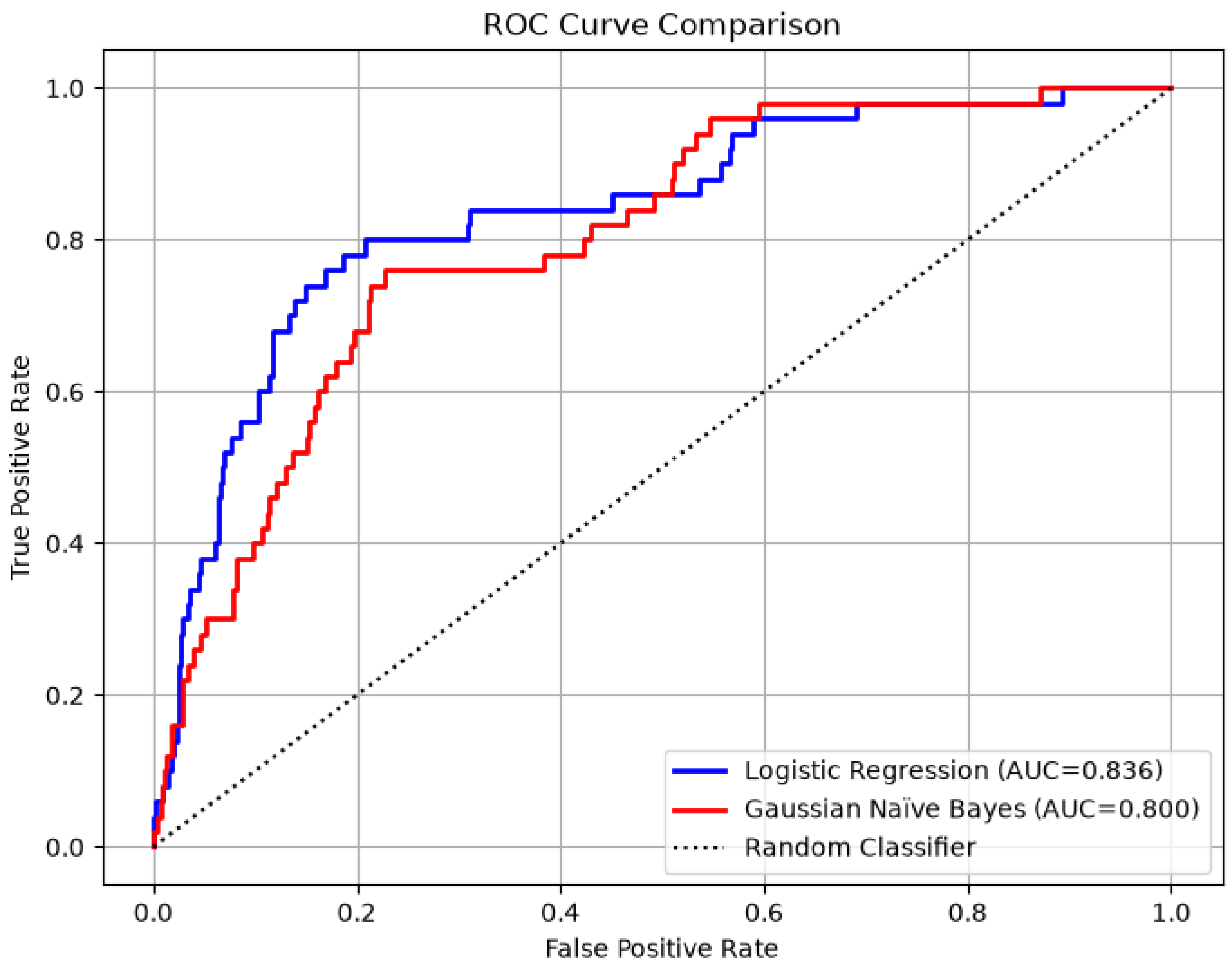


Figure 3: ROC curves for Logistic Regression (AUC = 0.836) and Naïve Bayes (AUC = 0.800).

The ROC Curve shows that the Logistic Regression outperformed Gaussian Naive Bayes in classifying stroke cases.

Logistic Regression achieved an Area Under the Curve (AUC) of 0.836 while the Gaussian obtained an AUC of 0.800. The Logistic Regression AUC indicates good discriminative ability as its ROC curve remains above the Gaussian Naives Bayes for most threshhold values indicating a higher True Positive Rate. It is more effective in identifying stroke patients than the Gaussian.

The Gaussian Naives Bayes (AUC =0.800) also performed well with an AUC at the lower end of the "good" classification range, however, its curve lies below the Logistic Regression indicating a slightly poorer discrimination between the two classes.

The Random Classifier which is the diagonal dashed line represent random guessing of (AUC =0.50). Both models performed better than the random confirming that they have learned meaningful patterns from the data. Overall, the ROC analysis indicates that the Logistic Rregression is the superior model achieving the highest AUC(0.836) compared to the Gaussian Naive Bayes (0.800). This suggests that the Logistic Regression has a better ability to distinguish between stroke and non-stroke cases and is therefore the preferred model for this stroke prediction task.

5. Discussion and Conclusion

5.1 Comparison of Model Performance

The four models exhibit sharply different trade-offs, driven largely by the severe class imbalance (only 4.9% positive cases):

- Logistic Regression and SVM (both tuned with class_weight = 'balanced') achieved the highest Recall (0.80 and 0.68 respectively), correctly identifying 40 of 50 stroke cases in the test set, but at the cost of low Precision (~0.11–0.13) and lower overall Accuracy (~71–73%), because they also generated a large number of false positives (260 and 262 respectively).
- Naïve Bayes struck the most balanced trade-off among the four models, achieving the highest F1-score (0.233), 86.5% accuracy, and a Recall of 0.42 — correctly flagging 21 of 50 stroke cases while producing far fewer false positives (109) than the balanced Logistic Regression/SVM models.
- KNN achieved the highest raw Accuracy (94.3%), but this is misleading: it identified only 1 of 50 stroke cases (Recall = 0.02), meaning it essentially predicts 'no stroke' for nearly every patient. On an imbalanced dataset like this, high accuracy alone does not indicate a clinically useful model. This model can score very high on accuracy while being clinically useless (as with KNN). F1-score and Recall are far more informative than accuracy alone for this problem.

5.2 Best-Performing Model

Choosing the 'best' model depends on the clinical objective:

- If the priority is maximizing F1-score (a balance of precision and recall) and overall accuracy, Naïve Bayes is the best performer among the four models tested (Accuracy = 0.865, F1 = 0.233).
- If the priority is maximizing Recall — i.e., minimizing missed stroke cases, which is typically the more clinically important error to avoid in a screening context — Logistic Regression (or SVM) is preferable, since both correctly identify 80% of actual stroke cases, at the cost of a higher false-positive rate that would require confirmatory follow-up testing.
- KNN is not recommended for this task in its current form, as its very low recall means it would fail to flag the vast majority of at-risk patients.

Considering that Logistic Regression also achieved the highest ROC-AUC (0.836) — reflecting the best overall ranking/discriminative ability across all thresholds — and offers transparent, clinically interpretable coefficients (odds ratios), it is recommended as the preferred model for a screening-oriented stroke-risk application, with Naïve Bayes as a strong, more precision-balanced alternative.

5.3 Strengths and Limitations of Each Technique

The comparative analysis demonstrates that each classification algorithm has unique strengths and weaknesses.

Logistic Regression provided interpretable probability estimates and served as a strong baseline classifier. Its simplicity makes it attractive for medical applications where interpretability is important.

Gaussian Naïve Bayes trained rapidly and produced probabilistic outputs. However, its assumption that predictor variables are conditionally independent may limit performance when healthcare variables are correlated.

Support Vector Machine effectively identified complex decision boundaries after feature scaling and hyperparameter optimisation. It generally performs well in high-dimensional feature spaces but is computationally more demanding.

K-Nearest Neighbours offered a simple, non-parametric approach capable of modelling non-linear relationships. Nevertheless, its prediction time increases with dataset size, and its performance is highly dependent on the selected number of neighbours.

The final choice of model should be based not only on overall accuracy but also on precision, recall, and F1-score, particularly because failing to identify an individual at risk of stroke (false negative) may have significant clinical consequences.

5.4 Recommendations

- For a stroke-screening tool where missing a true stroke case carries a high cost, deploy a balanced Logistic Regression model with a threshold selected to prioritize Recall, and use its output probabilities to triage patients for further clinical review.
- Naïve Bayes offers a strong, low-maintenance alternative when a better precision/recall balance is desired, e.g., for a resource-constrained clinical setting.
- Future work should address the class imbalance more directly — e.g., through oversampling techniques such as SMOTE, cost-sensitive learning, or ensemble methods (Random Forest, Gradient Boosting) — and should evaluate models using cross-validated ROC-AUC rather than a single train-test split to obtain more robust performance estimates.
- Given the modest AUC values (0.80–0.84), collecting additional clinical predictors (e.g., family history, prior transient ischemic attacks, medication use) may further improve model discrimination beyond what is achievable with the current feature set.